

Use the equation below to answer questions 1 through 3.

$$y = a(x - 1)(x + 3)$$

- 1. Write the x -intercepts of the quadratic equation.
- 2. Solve for a , given that $(-2, -6)$ is a point on the quadratic's graph.
- 3. Using the value of a found in question 2, write the quadratic equation in factored form.
- 4. Rewrite the quadratic equation in standard form.

The x -intercepts and a point on the parabola are given for several quadratic functions. For each function, show how to find the value of a and write each quadratic in factored form.

	x -intercepts	Point	Steps to Find " a "	Quadratic Equation in Factored Form
5.	(1,0) and (4,0)	(-2,6)		

	x-intercepts	Point	Steps to Find “a”	Quadratic Equation in Factored Form
6.	$(-5, 0)$ and $(-2, 0)$	$(4, -3)$		
7.	$(\frac{1}{2}, 0)$ and $(3, 0)$	$(-1, \frac{3}{2})$		
8.	$(0, 0)$ and $(5, 0)$	$(1, -24)$		
9.	$(-3, 0)$ and $(-3, 0)$	$(-2, 2)$		
10.	$(2, 0)$ and $(-2, 0)$	$(4, 3)$		